

microGPT – 1Page Formula Cheat Sheet

Embedding: $e_i = E[\text{token}_i]$

Positional Add: $x_i = e_i + p_i$

Attention: $\text{softmax}((QK^T)/\sqrt{d_k}) V$

Softmax: $\exp(z_i) / \sum \exp(z_j)$

LayerNorm: $\gamma ((x-\mu)/\sqrt{(\sigma^2+\epsilon)}) + \beta$

FFN: $\max(0, xW+b) W + b$

Residual: $y = x + F(x)$

Loss: $L = -\log(p_{\text{correct}})$

Update: $\theta \leftarrow \theta - \eta \nabla L$